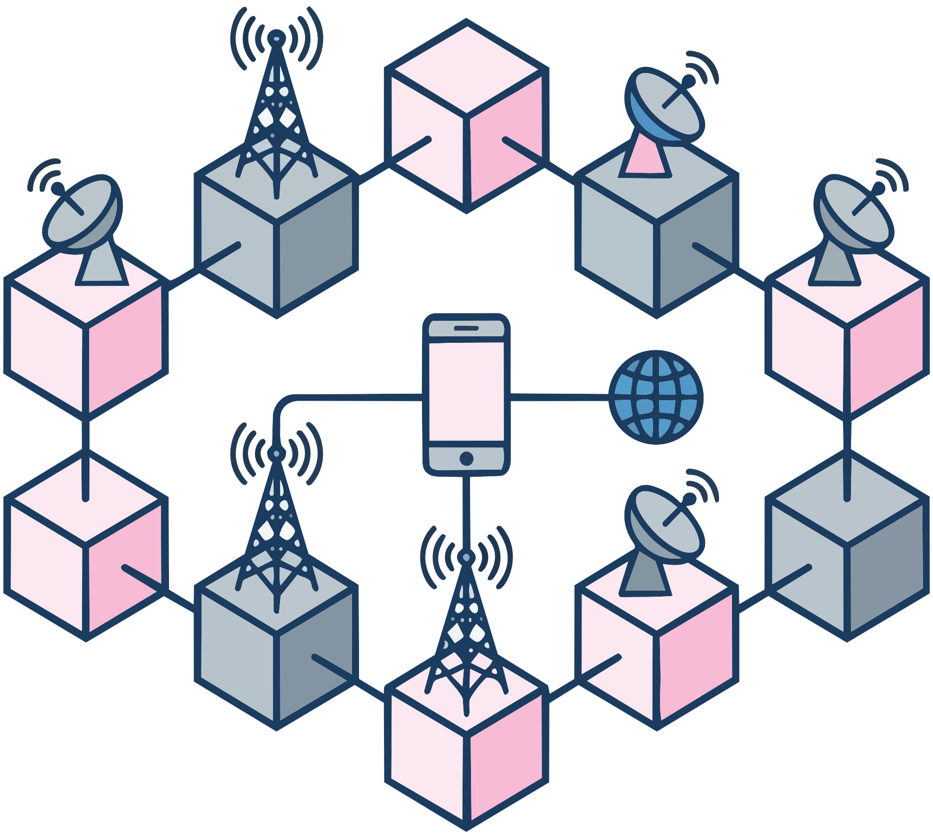
Blockchain: The Tech for Telecom

Blockchain applications are moving beyond finance and money to radio waves, IoT, and a range of telecommunications.



Blockchain technology is generally associated with banking, finance and monetary transactions. However, this technology can now be used for radio waves, Internet of Things (IoT), RFID tags, laser communications, light fidelity (LiFi), airwaves and many similar communication scenarios. Blockchain can be used for dynamic spectrum allocation and real-time analytics on radio waves and IoT signals for multiple industrial, government, corporate as well as social applications.

Key applications and use cases of blockchain in radio frequency (RF) and telecom are:

Dynamic spectrum management (DSM)*:* In classical industrial applications, the radio and telecom spectrum is allocated and distributed by government bodies in fixed blocks (FM radio/LTE/4G/5G bands). This may be inefficient because of the huge traffic and load involved. A blockchain-based solution provides a decentralised ledger without any centralised controller, which acts as a real-time database for dynamic spectrum evaluation, distribution and spectrum access systems (SAS).

Decentralised wireless networks (DeWi) and decentralised antenna design: This is a high performance and real-world application with the integration of decentralised antenna design. Rather than using proof of work (as in Bitcoin), these networks can use radio frequency (RF) and IoT signals to analyse in a dynamic format that a particular hotspot is providing good coverage in a specific location or geographic area.